

DecisionHub

1. Title

DecisionHub: Collaborative Decision-Making & Community Polling Platform

2. Objective

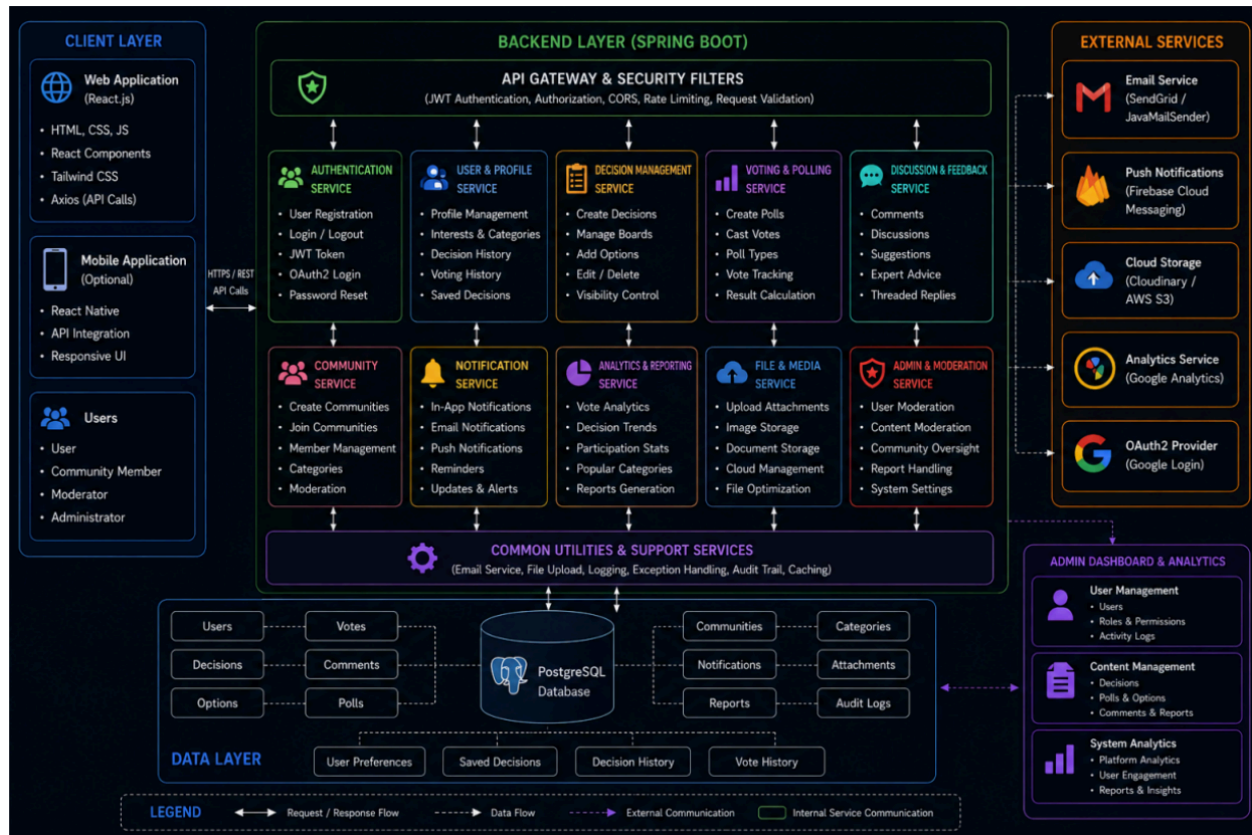
Build a full-stack web application that enables users to create decision boards, compare multiple options, invite friends or communities to vote, gather feedback, analyze pros and cons, and make informed decisions using collective intelligence.

The platform supports structured decision analysis, community discussions, voting workflows, polling systems, analytics dashboards, and collaborative decision-making features.

Outcomes

- Developed and deployed a full-stack React.js + Spring Boot application.
 - Implemented secure authentication using JWT and OAuth2.
 - Built decision board creation and management functionality.
 - Implemented voting and polling systems.
 - Added option comparison and pros/cons evaluation modules.
 - Created decision analytics dashboards and reporting systems.
 - Implemented community collaboration and discussion features.
 - Developed notification and engagement workflows.
 - Deployed frontend and backend using PostgreSQL in production.
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3. Architecture Diagram



4. Modules to be Implemented

1. User Authentication & Role-Based Access

- JWT Authentication
- OAuth2 Login
- Password Reset
- Profile Management

Roles

- User
- Community Moderator
- Administrator

2. User Profile Management

- Profile creation
- Interests and categories
- Decision history
- Voting history
- Saved decisions
- Account settings

3. Decision Board Management

- Create decision boards
- Add multiple options
- Edit decision boards
- Delete decision boards
- Public decision boards
- Private decision boards

Example Decisions

- MBA vs Job
- iPhone vs Samsung
- Goa vs Bali
- Startup vs Corporate Job
- Remote vs Office Work

4. Option Comparison System

- Add comparison options
- Feature comparison
- Pros and cons listing
- Score allocation
- Option ranking

Comparison Factors

- Cost
- Benefits
- Risk
- Time
- Convenience
- User-defined criteria

5. Voting & Polling System

- Public voting
- Private voting
- Anonymous voting
- Poll creation
- Vote tracking
- Result visualization

Poll Types

- Single Choice Polls
- Multiple Choice Polls
- Rating-Based Polls

6. Discussion & Feedback System

- Community discussions
- Comments
- Suggestions
- Advice sharing
- Expert recommendations
- Threaded conversations

7. Decision Analytics Dashboard

User Dashboard

- Active decisions
- Voting participation
- Decision outcomes
- Popular categories
- Community activity

Analytics

- Vote distribution
- Decision trends
- Option popularity
- Participation statistics

8. Notification System

- Voting reminders
- New comments notifications
- Poll completion alerts
- Community invitations
- Decision updates
- System notifications

9. Community Management

- Create communities
- Join communities
- Category-based groups
- Community moderation

- Member management

Categories

- Career
- Education
- Technology
- Travel
- Finance
- Lifestyle

10. Reports & Export System

- Decision reports
- Poll results
- Voting analytics
- Community reports
- PDF export
- Excel export

11. Final Integration, Testing & Deployment

- Frontend and backend integration
- API validation and testing
- End-to-end workflow testing
- Security testing
- Performance optimization
- Docker containerization
- Production deployment
- Monitoring and logging
- Documentation and user guides

5. Week-wise Module Implementation and High-Level Requirements

Milestone 1: Week 1 & 2 — Requirements, Database Design & Backend Setup

Tasks

- Define project scope and user roles.
- Design database schema:
 - Users
 - Decisions
 - Options
 - Votes
 - Comments
 - Communities
 - Notifications
 - Reports
- Initialize Spring Boot project.
- Configure PostgreSQL/MySQL.
- Implement JWT authentication.
- Create React frontend skeleton.

Outcomes

- Backend and frontend architecture setup completed.
- Authentication flow functional.
- Database schema finalized.
- Role-based access configured.

Milestone 2: Week 3 & 4 — Decision & Voting Management

Tasks

- Implement decision board APIs.
- Build voting and polling systems.
- Develop option comparison functionality.
- Create community pages.
- Implement result tracking.

Outcomes

- Decision management operational.
- Voting workflow functional.
- Community participation completed.

Milestone 3: Week 5 & 6 — Discussions & Notifications

Tasks

- Implement discussion forums.
- Build notification service.
- Develop analytics APIs.
- Create feedback workflows.
- Add moderation features.

Outcomes

- Discussion management operational.
- Notification system functional.
- Analytics modules completed.

Milestone 4: Week 7 & 8 — Analytics, Testing & Deployment

Tasks

- Build analytics dashboard.
- Add charts and reports.
- Implement testing and validations.
- Deploy frontend/backend using AWS, Render, or Vercel.
- Configure SSL and production environment.

Outcomes

- Fully deployed production-ready application.
 - Analytics and reporting systems operational.
 - Complete end-to-end decision-making workflow demonstrable.
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6. Evaluation Criteria

Milestone 1 (Week 2)

- Spring Boot project setup completed.
- JWT authentication implemented.
- Database schema finalized.
- Frontend authentication flow working.

Milestone 2 (Week 4)

- Decision management operational.
- Voting system functional.
- Comparison workflow implemented.

Milestone 3 (Week 6)

- Discussion system operational.
- Notification workflow functional.

- Analytics modules completed.

Milestone 4 (Week 8)

- Fully deployed frontend and backend.
 - Dashboard and reporting systems operational.
 - End-to-end decision-making workflow demonstrated.
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7. Tools & Tech Stack

Programming Language

Backend

- Java
- Spring Boot

Frontend

- JavaScript
- React.js

Database

Local Development

- MySQL

Production

- PostgreSQL

Authentication

- Spring Security
- JWT Authentication
- OAuth2 (Google Login)

Libraries & Frameworks

Backend

- Spring Boot
- Spring Security
- Spring Data JPA
- Hibernate
- Maven

Frontend

- React
- React Router
- Axios
- Tailwind CSS
- Context API
- Chart.js

Notifications

- Firebase Cloud Messaging
- JavaMailSender

Testing

- JUnit
- Mockito
- Postman
- React Testing Library

Dev & Deployment Tools

- IntelliJ IDEA

- VS Code
 - Git & GitHub
 - Docker
 - Docker Compose
 - GitHub Actions
 - AWS / Render / Railway
 - Postman
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8. Performance Metrics

Decision Management Metrics

- Decision board creation success rate
- Community participation rate
- Poll completion rate
- User engagement score

Voting & Analytics Metrics

- Voting accuracy
- Poll participation rate
- Option comparison usage
- Analytics report generation success rate

System Performance Metrics

- API response time
 - Dashboard loading speed
 - Notification delivery success rate
 - Concurrent user handling capacity
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9. Example Quantitative Goals

Decision Boards

Enable users to create and manage structured decision boards effectively.

Voting & Polling

Support high participation rates and accurate vote collection.

Community Collaboration

Encourage active discussions and community engagement.

Analytics & Reporting

Provide actionable insights through decision analytics dashboards.

Platform Performance

Support large-scale concurrent participation while maintaining stable system performance.